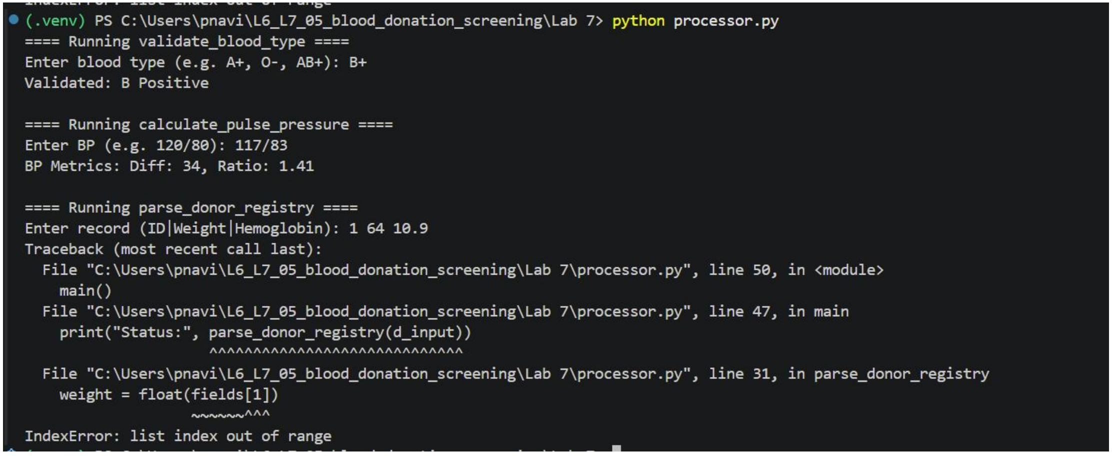
LIKITH
PES1UG24CS250
SECTION E

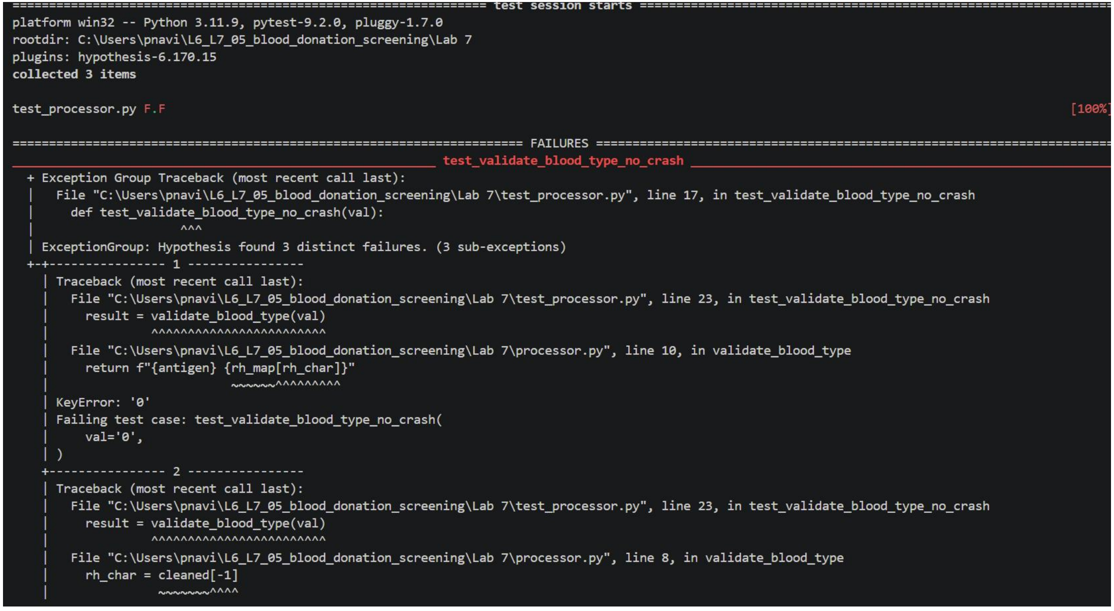
My analysis:

Ran processor.py

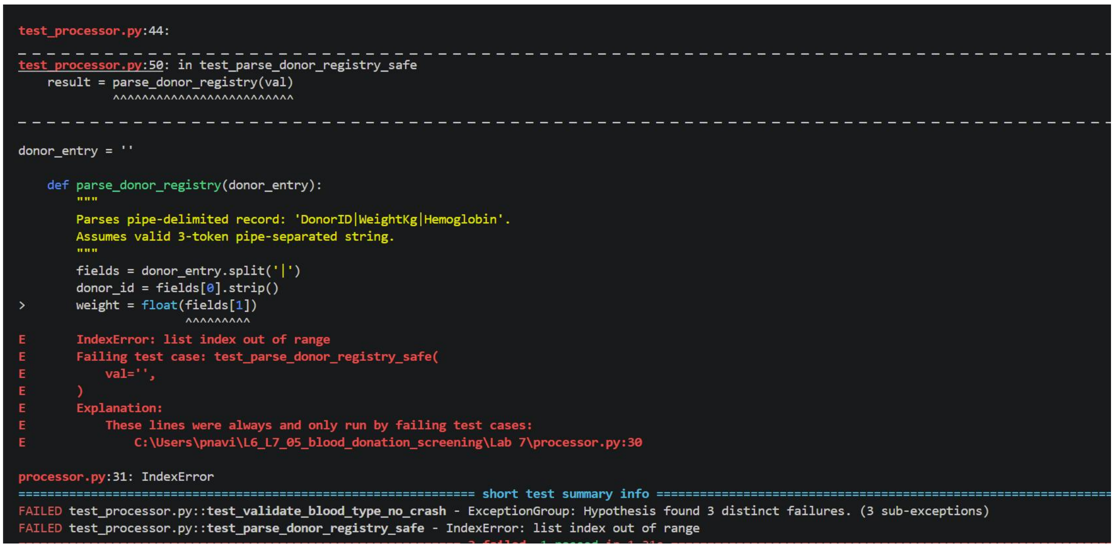


```
(.venv) PS C:\Users\pnavi\L6_L7_05_blood_donation_screening\Lab 7> python processor.py
==== Running validate_blood_type ====
Enter blood type (e.g. A+, O-, AB+): B+
Validated: B Positive

==== Running calculate_pulse_pressure ====
Enter BP (e.g. 120/80): 117/83
BP Metrics: Diff: 34, Ratio: 1.41

==== Running parse_donor_registry ====
Enter record (ID|Weight|Hemoglobin): 1 64 10.9
Traceback (most recent call last):
  File "C:\Users\pnavi\L6_L7_05_blood_donation_screening\Lab 7\processor.py", line 50, in <module>
    main()
  File "C:\Users\pnavi\L6_L7_05_blood_donation_screening\Lab 7\processor.py", line 47, in main
    print("Status:", parse_donor_registry(d_input))
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\pnavi\L6_L7_05_blood_donation_screening\Lab 7\processor.py", line 31, in parse_donor_registry
    weight = float(fields[1])
                   ~~~~~~^^^
IndexError: list index out of range
```

Ran test_processor.py using pytest

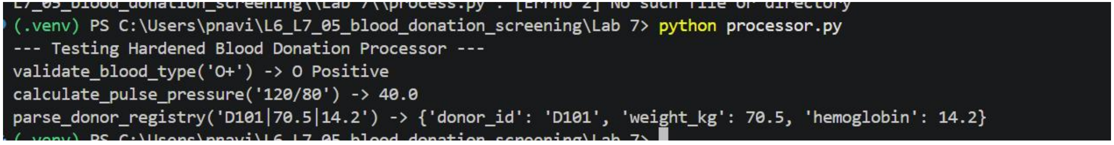


```
============================================================ test session starts =============================================================
platform win32 -- Python 3.11.9, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\pnavi\L6_L7_05_blood_donation_screening\Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_processor.py F.F                                                                                                                   [100%]

================================================================== FAILURES ==================================================================
_________________________________________________________ test_validate_blood_type_no_crash _________________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\pnavi\L6_L7_05_blood_donation_screening\Lab 7\test_processor.py", line 17, in test_validate_blood_type_no_crash
  |     def test_validate_blood_type_no_crash(val):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\pnavi\L6_L7_05_blood_donation_screening\Lab 7\test_processor.py", line 23, in test_validate_blood_type_no_crash
    |     result = validate_blood_type(val)
    |              ^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\pnavi\L6_L7_05_blood_donation_screening\Lab 7\processor.py", line 10, in validate_blood_type
    |     return f"{antigen} {rh_map[rh_char]}"
    |                         ~~~~~~^^^^^^^^^
    | KeyError: '0'
    | Failing test case: test_validate_blood_type_no_crash(
    |     val='0',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\pnavi\L6_L7_05_blood_donation_screening\Lab 7\test_processor.py", line 23, in test_validate_blood_type_no_crash
    |     result = validate_blood_type(val)
    |              ^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\pnavi\L6_L7_05_blood_donation_screening\Lab 7\processor.py", line 8, in validate_blood_type
    |     rh_char = cleaned[-1]
    |               ~~~~~~~^^^^
    |
```



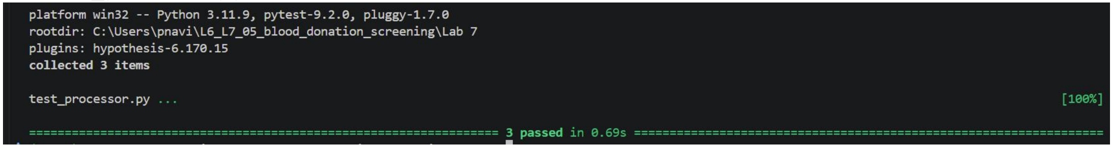
```
test_processor.py:44:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
test_processor.py:50: in test_parse_donor_registry_safe
    result = parse_donor_registry(val)
             ^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

donor_entry = ''

    def parse_donor_registry(donor_entry):
        """
        Parses pipe-delimited record: 'DonorID|WeightKg|Hemoglobin'.
        Assumes valid 3-token pipe-separated string.
        """
        fields = donor_entry.split('|')
        donor_id = fields[0].strip()
>       weight = float(fields[1])
                       ^^^^^^^^^
E       IndexError: list index out of range
E       Failing test case: test_parse_donor_registry_safe(
E           val='',
E       )
E       Explanation:
E           These lines were always and only run by failing test cases:
E               C:\Users\pnavi\L6_L7_05_blood_donation_screening\Lab 7\processor.py:30

processor.py:31: IndexError
============================================================== short test summary info ==============================================================
FAILED test_processor.py::test_validate_blood_type_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_parse_donor_registry_safe - IndexError: list index out of range
```

Changed processor.py file with new code with hardcoded values



```
(.venv) PS C:\Users\pnavi\L6_L7_05_blood_donation_screening\Lab 7> python processor.py
--- Testing Hardened Blood Donation Processor ---
validate_blood_type('O+') -> O Positive
calculate_pulse_pressure('120/80') -> 40.0
parse_donor_registry('D101|70.5|14.2') -> {'donor_id': 'D101', 'weight_kg': 70.5, 'hemoglobin': 14.2}
```

Ran test_processor.py file



```
platform win32 -- Python 3.11.9, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\pnavi\L6_L7_05_blood_donation_screening\Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_processor.py ...                                                                                [100%]

================================================================ 3 passed in 0.69s ================================================================
```